ROLES & PERMISSIONS MODULE

Technical Design Document | v1.3

ASP.NET Core MVC • SQL Server 2019+ • School ERP • RBAC Architecture

Project Name	Roles & Permissions Module — School ERP
Application	ASP.NET Core MVC with SQL Server
Version	1.3
Prepared By	Development Team
Reviewed By	Technical Lead
Approved By	Project Manager
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Revision History: v1.0 — Initial Technical Design Document | v1.1 — Diagrams, index details, multilingual datatypes | v1.2 — ER diagram, stored procedures, views, triggers | v1.3 — School ERP scope, organisation hierarchy, OTP authentication, per-table history, expanded permission actions, session management, department-driven menu

01 Document Control

Version	Date	Prepared By	Remarks
1.0	20-Mar-2026	Dev Team	Initial release of Technical Design Document
1.1	23-Mar-2026	Dev Team	Diagrams, index details, multilingual datatypes
1.2	23-Mar-2026	Dev Team	ER diagram, stored procedures, views, triggers
1.3	25-Mar-2026	Dev Team	School ERP scope, organisation hierarchy, OTP auth, per-table history tables, expanded permission actions, academic session, back-date control, department-driven menu

02 Purpose and Scope

This document describes how the Roles & Permissions module is designed, implemented, secured, deployed, and maintained for the School Management ERP system. It is intended for developers, testers, infrastructure teams, support engineers, and future handover stakeholders.

In scope: application architecture, code structure, module design, database schema with all objects, integrations, configuration, security, deployment, logging, backups, and operational support.

> Separation of Documents: This is the TECHNICAL document only. The FUNCTIONAL document — covering Business Process, User Requirements, Screen Behaviour, Approval Flow, and Business-Side Reports — is a separate document.

03 Solution Overview

The Roles & Permissions module provides secure, flexible Role-Based Access Control (RBAC) built on ASP.NET Core MVC with ADO.NET and SQL Server, extended for a multi-school ERP context. The module supports multi-role users, permission inheritance, soft deletes, per-table audit history, multilingual data storage via NVARCHAR fields, and academic session-scoped data management.

Component	Purpose	Implementation
Presentation Layer	UI, routing, validation, view rendering	ASP.NET Core MVC controllers, Razor views, ViewModels
Business Layer	Business rules and orchestration	Services, validators, transaction handling
Data Layer	Read/write data to SQL Server	ADO.NET, stored procedures, parameterized queries — NO ORM
Security Layer	JWT auth, RBAC enforcement, encryption	JWT Bearer, AES-256, claims-based authorization
OTP Layer	Two-factor authentication per user setting	OTP generation + validation service; configurable at user creation
Org Hierarchy Layer	Multi-school / multi-branch data scoping	MainAccountID + ParentID on all master tables; session-scoped queries

04 Technology Stack

Layer	Technology	Remarks
Frontend	HTML5, CSS3, Bootstrap, JavaScript, jQuery	Responsive UI and client-side enhancements
Backend	ASP.NET Core MVC (.NET 8)	Server-side rendering, routing, middleware
Language	C#	Business logic and controller/service code
Database	Microsoft SQL Server 2019+	Transactional relational database
Data Access	ADO.NET	Direct SQL via stored procedures — no ORM, no ad-hoc SQL
Authentication	JWT Bearer + OTP (2FA)	Stateless JWT; OTP configurable per user at creation
Authorisation	Claims / Policy-based RBAC	Role and permission-based access control
Logging	Serilog / NLog	Structured logs and diagnostics
Audit History	Per-table history tables	Each master table has a dedicated _History table
Hosting	IIS / Windows Server / Azure App Service	Production deployment model

06 User Roles and Access Matrix

The module defines four primary actor types for the School ERP. Every access decision traces back through the chain: Super Admin creates Roles, assigns Permissions to Roles, creates Users, and assigns Roles to Users.

6.1 Super Admin

Attribute	Details
Who	Platform-level administrator — system owner or IT administrator.
What They Do	Create, edit, activate, deactivate Roles; assign/remove Permissions from Roles; create companies/branches; create, edit, lock, unlock, deactivate User accounts; assign one or multiple Roles to any User; view full audit logs and access history; manage all module-level permissions across the application. Any user with the relevant menu access can also perform these actions.
Cannot Do	Cannot assign permissions directly to users — must always go through a Role.

6.2 Admin (Role-based)

Attribute	Details
Who	Department-level or module-level admin created and configured by Super Admin.
What They Do	Manage users within their assigned scope or department; perform Create / Edit / View on permitted modules; view audit logs within their scope.
Limitations	Cannot create Roles or modify Role-Permission mappings. All access is governed by their assigned Role.

6.3 Staff / End User

Attribute	Details
Who	Regular application user — employee, operator, or any end user.
What They Do	Access only modules and actions granted by their assigned Role(s). Actions include: Add / Edit / Delete / Export / Print / Cancel based on their specific permissions. Can have multiple roles — permissions are combined (UNION).
Limitations	Cannot manage roles, permissions, or other users. Access is read from role-permission mapping on every request.

6.4 Teacher

Attribute	Details
Who	Teaching staff — specific actor type in the School ERP context.
What They Do	Access class, student, attendance, exam, and grade modules as permitted by their assigned Role.

Limitations	Cannot manage roles, permissions, or administrative users. School/branch-scoped access only.

6.5 Role Structure

Role Name	Module Access	Actions Allowed	Typical User
Super Admin	All Modules	All Actions	System Owner / IT Head
HR Manager	Users, Roles	View, Add, Edit	HR Department Head
Finance Staff	Reports, Fee	View, Add, Export, Print	Accounts Officer
Viewer	Dashboard, Reports	View only	Management / Auditor
Teacher	Class, Attendance, Exam	View, Add, Edit	Teaching Staff

06 Switch Profile

When a user is assigned multiple Roles across different modules or contexts, the system displays a Switch Profile icon in the navigation header. This enables the user to switch their active working context without logging out and back in. The icon appears only when the authenticated user holds two or more role or profile assignments.

Display Rule

The Switch Profile icon appears in the navigation header only when the logged-in user has two or more active role or profile assignments. For users with a single role, the icon is hidden and the user operates directly within their default context.

6.5.1 Real-World Example — Multi-Context User (School ERP)

Consider Rajesh Mehta — an employee in a School ERP organisation who holds multiple roles: he is an Accountant, a Fee Collector and an HR Manager within the ERP system, and is also a parent of two children enrolled in the same school. After login, the Switch Profile dropdown presents all four contexts:

#	Profile / Context	Role(s)	Module Access Granted
1	ERP Login (Organisation Portal)	Accountant, Fee Collector, HR Manager	Accounts, Fee Management, HR Module, Reports
2	Employee Profile (Self-Service Portal)	Employee (Self)	Payslips, Leave Requests, Attendance, Personal Details
3	Student 1 – Aryan Mehta (Parent Portal)	Parent / Guardian	Attendance, Exam Results, Fee Dues, School Notices
4	Student 2 – Priya Mehta (Parent Portal)	Parent / Guardian	Attendance, Exam Results, Fee Dues, School Notices

6.5.2 Technical Flow — Switch Profile

Step	Detail
1	On login, the system queries tbl_map_user_roles to count distinct profile contexts for the user. If count ≥ 2, the Switch Profile icon is rendered in the navigation header.
2	Clicking the icon opens a dropdown listing all available profile contexts with their role labels. Single-role users see no icon.
3	On profile selection, active JWT claims are refreshed with the selected RoleIDs and the UI reloads permitted modules for that context.
4	The switch event is logged to the history table with ActionType = PROFILE_SWITCH, capturing the previous and new active profile for audit traceability.

> Key Rule: Switch Profile does not bypass RBAC. Each profile context enforces its own independent role-permission set. Permissions from one profile context are never carried into another. All access decisions continue to be evaluated against vw_UserEffectivePerms for the active context.

06A Organisation Hierarchy

The system supports a multi-school, multi-branch structure to manage multiple schools under one platform.

6A.1 Hierarchy Structure

Level	Entity	Description	Key Fields
Level 1	Group (Head Office)	Top-level parent organisation	MainAccountID (root), ParentID = NULL
Level 2	Sub-groups (Schools)	Individual schools under the group	MainAccountID, ParentID = Group ID
Level 3	Users	Users created with single or multiple school access	UserID, MainAccountID, SchoolIDs[]

6A.2 Data Scoping Rules

- All master records carry MainAccountID — ensures data isolation between groups.
- Users can be granted access to one or multiple schools/branches.
- All data queries are scoped by the active school context (OrgID) selected at login.
- Reports can be generated school-wise or at group level, depending on the user's role.

6A.3 New Tables

Table	Purpose	Key Columns
tbl_mst_organisations	Stores group and school records	OrgID, OrgName, OrgType (Group/School), MainAccountID, ParentID
tbl_map_user_schools	Maps users to accessible schools/branches	UserID, OrgID, AssignedBy, AssignedOn

06B Permission Actions

The system supports the following permission action types. Each module-action pair is defined as a distinct permission record in tbl_mst_permissions and can be independently assigned to any Role.

Action	Description
View	Read-only access to module data
Add	Create new records within the module
Edit	Modify existing records
Delete	Soft delete — sets IsActive=0; no hard delete permitted
Export	Export data to Excel/CSV/PDF — controlled separately from View
Print	Print reports or records — controlled separately from View
Cancel	Cancel a transaction or entry — applicable to operational modules

Note: The Action column in tbl_mst_permissions (VARCHAR(20)) accommodates all values above. Seed data must include Export, Print, and Cancel entries for each applicable module.

06C Session Management and Back-date Control

6C.1 Academic Session

All operational data is scoped to an academic session (SessionID). Users select the active school and session context at login if multiple options exist.

Aspect	Detail
SessionID	Present on all operational tables. Ensures data is partitioned by academic year/session.
School Selection	On login, if the user has access to multiple schools, they select the active school/branch. The active OrgID is stored in JWT claims.
Reports	Available school-wise and at group (head-office) level based on the user's role.

6C.2 Back-date Entry Control

Aspect	Detail
Scope	User-specific setting — some users may be permitted to enter data for past dates; others are restricted to the current date.
DB Column	AllowBackdate BIT in tbl_mst_users. 1 = back-date entry permitted; 0 = restricted to current date.
Enforcement	Service layer validates the entry date against the AllowBackdate flag before processing any dated transaction.

06D Menu Assignment and Department Configuration

6D.1 Department-Driven Menu Access

Departments define which menu items and functional areas a user can access. Menu visibility is dynamic and driven by the user's role and department assignment.

Feature	Detail
Dynamic Menu	Menu items are rendered based on the role-permission mapping. Users see only what their active role permits.
Menu Alignment / Display Control	Administrators can control the order and visibility of menu items per role.
Bulk Permission Assignment	Permissions can be assigned in bulk at the department level and inherited by all users in that department/role.
Department Table	tbl_mst_departments: DeptID, DeptName, MainAccountID, IsActive. Linked to users via tbl_map_user_departments.

11 Database Design and Data Dictionary

Multilingual Datatype Note: All text fields that may contain multilingual content use NVARCHAR (Unicode / UTF-16) to support Arabic, Hindi, Chinese, and all other scripts. System codes and fixed enum values use VARCHAR. All timestamps use DATETIME2(0) in UTC.

11.1 Tables Summary

Table Name	Type	Primary Key	Purpose
tbl_mst_roles	Master	RoleID (INT)	Stores all role definitions
tbl_mst_permissions	Master	PermissionID (INT)	Stores module-action permission definitions
tbl_map_role_permissions	Mapping	RoleID + PermissionID	Maps roles to permissions (M:M)
tbl_mst_users	Master	UserID (INT)	Stores user master records
tbl_map_user_roles	Mapping	UserID + RoleID	Maps users to roles (M:M)
tbl_aud_log	Audit	AuditID (BIGINT)	Tracks system-level events: login, logout, profile switch
tbl_mst_organisations	Master	OrgID (INT)	Group and school hierarchy records
tbl_map_user_schools	Mapping	UserID + OrgID	Maps users to accessible schools/branches
tbl_mst_departments	Master	DeptID (INT)	Department definitions — drives menu access
tbl_map_user_departments	Mapping	UserID + DeptID	Maps users to departments
tbl_mst_*_History (per table)	History	HistoryID (BIGINT)	Full row snapshot per change: UpdateDeleteFlag, DoneBy, DoneOn

11.2 tbl_mst_roles

Column	Data Type	Null	Description
RoleID	INT IDENTITY(1,1) PK	NO	System-generated unique role identifier
RoleName	NVARCHAR(150)	NO	Multilingual role name
RoleDesc	NVARCHAR(500)	YES	Multilingual description of the role
IsActive	BIT	NO	1=Active, 0=Inactive (soft delete)
CreatedBy	INT FK	NO	UserID of the user who created this record
CreatedOn	DATETIME2(0)	NO	UTC timestamp of record creation
ModifiedBy	INT FK	YES	UserID of last modifier

ModifiedOn	DATETIME2(0)	YES	UTC timestamp of last modification

11.3 tbl_mst_permissions

Column	Data Type	Null	Description
PermissionID	INT IDENTITY(1,1) PK	NO	Unique permission identifier
ModuleName	NVARCHAR(200)	NO	Module name — multilingual
ModuleCode	VARCHAR(50)	NO	System code (ASCII) e.g. FEE_MGMT
Action	VARCHAR(20)	NO	Action type: View / Add / Edit / Delete / Export / Print / Cancel
DisplayLabel	NVARCHAR(200)	YES	Multilingual UI label
IsActive	BIT	NO	1=Active, 0=Inactive
CreatedBy	INT FK	NO	UserID of creator
CreatedOn	DATETIME2(0)	NO	UTC creation timestamp

11.4 tbl_map_role_permissions

Primary Key: Composite (RoleID, PermissionID) — prevents duplicate permission assignments to the same role.

Column	Data Type	Null	Description
RoleID	INT FK	NO	Foreign key to tbl_mst_roles.RoleID
PermissionID	INT FK	NO	Foreign key to tbl_mst_permissions.PermissionID
AssignedBy	INT FK	NO	UserID of Super Admin who assigned this mapping
AssignedOn	DATETIME2(0)	NO	UTC timestamp of assignment

11.5 tbl_mst_users

Column	Data Type	Null	Description
UserID	INT IDENTITY(1,1) PK	NO	System-generated unique user identifier
FullName	NVARCHAR(300)	NO	Full name — multilingual
Username	NVARCHAR(150)	NO	Login username — unique
Email	NVARCHAR(250)	NO	Email address — unique constraint enforced
PasswordHash	NVARCHAR(500)	NO	AES-256 encrypted password hash

PasswordSalt	NVARCHAR(200)	NO	Per-user salt for password hashing
PhoneNo	NVARCHAR(30)	YES	Contact phone number
IsOTPEnabled	BIT	NO	1 = OTP required on login; 0 = Password only
OTPSecret	NVARCHAR(200)	YES	Encrypted OTP seed/secret
OTPExpiry	DATETIME2(0)	YES	UTC expiry of the current issued OTP
AllowBackdate	BIT	NO	1 = back-date entry permitted; 0 = restricted to current date
IsActive	BIT	NO	1=Active, 0=Inactive (soft delete)
IsLocked	BIT	NO	1=Account locked after failed login attempts
FailedAttempts	TINYINT	NO	Count of consecutive failed login attempts
LastLoginOn	DATETIME2(0)	YES	UTC timestamp of last successful login
CreatedBy	INT FK	NO	UserID of creator
CreatedOn	DATETIME2(0)	NO	UTC creation timestamp
ModifiedBy	INT FK	YES	UserID of last modifier
ModifiedOn	DATETIME2(0)	YES	UTC modification timestamp

11.6 tbl_map_user_roles

Primary Key: Composite (UserID, RoleID) — prevents duplicate role assignments for the same user.

Column	Data Type	Null	Description
UserID	INT FK	NO	Foreign key to tbl_mst_users.UserID
RoleID	INT FK	NO	Foreign key to tbl_mst_roles.RoleID
AssignedBy	INT FK	NO	UserID of Super Admin who made this assignment
AssignedOn	DATETIME2(0)	NO	UTC timestamp of assignment

11.7 tbl_aud_log (Audit Log — System Events)

tbl_aud_log is write-heavy. Implement a quarterly archive job to tbl_aud_log_archive for records older than 12 months. Never apply DELETE cascades to this table.

Column	Data Type	Null	Description
AuditID	BIGINT IDENTITY(1,1) PK	NO	Unique audit record identifier
UserID	INT FK	NO	UserID of the user who performed the action
ModuleName	NVARCHAR(200)	NO	Module where the action occurred

ActionType	VARCHAR(20)	NO	LOGIN / LOGOUT / PROFILE_SWITCH / LOCK / UNLOCK
RecordID	INT	YES	ID of the affected record
IPAddress	NVARCHAR(50)	YES	Client IP address
DoneOn	DATETIME2(0)	NO	UTC timestamp of the action
Remarks	NVARCHAR(500)	YES	Free-text remarks or reason

11.8 Per-Table History Tables

Each master table has a corresponding _History table that captures full row snapshots on every UPDATE and DELETE. The history table mirrors all columns of the master table plus:

Column	Data Type	Null	Description
HistoryID	BIGINT IDENTITY(1,1) PK	NO	Unique history record identifier
[All columns of master table]	Same as source	—	Full snapshot of the row at time of change
UpdateDeleteFlag	CHAR(1)	NO	U = Update, D = Delete
DoneBy	INT FK	NO	UserID of the user who performed the action
DoneOn	DATETIME2(0)	NO	UTC timestamp of the change

History Tables

History Table	Source Master Table
tbl_mst_roles_History	tbl_mst_roles
tbl_mst_permissions_History	tbl_mst_permissions
tbl_mst_users_History	tbl_mst_users
tbl_mst_organisations_History	tbl_mst_organisations
tbl_map_role_permissions_History	tbl_map_role_permissions
tbl_map_user_roles_History	tbl_map_user_roles

14 Stored Procedures

All database operations are performed via stored procedures. Naming convention: sp_Module_Action. No ad-hoc SQL is permitted in the application layer.

14.1 Role Management

Procedure Name	Action	Description
sp_Roles_Create	INSERT	Creates a new role — validates unique name before insert
sp_Roles_Update	UPDATE	Updates role name/description — validates uniqueness, writes history
sp_Roles_Deactivate	UPDATE	Sets IsActive=0 — checks no active users before deactivating
sp_Roles_GetAll	SELECT	Returns all roles with active/inactive filter support
sp_Roles_GetByID	SELECT	Returns a single role record by RoleID

14.2 Permission Management

Procedure Name	Action	Description
sp_Permissions_Create	INSERT	Creates a permission — validates unique ModuleCode+Action
sp_Permissions_GetAll	SELECT	Returns all permissions, optionally filtered by module
sp_RolePerms_Assign	INSERT	Assigns a permission to a role — ignores if already assigned
sp_RolePerms_Revoke	DELETE	Removes a permission from a role
sp_RolePerms_GetByRole	SELECT	Returns all permissions assigned to a given RoleID

14.3 User Management

Procedure Name	Action	Description
sp_Users_Create	INSERT	Creates a new user — validates unique Username and Email
sp_Users_Update	UPDATE	Updates user details — writes history record
sp_Users_Deactivate	UPDATE	Sets IsActive=0 for user — soft delete
sp_Users_Lock	UPDATE	Sets IsLocked=1 after max failed login attempts

sp_Users_Unlock	UPDATE	Sets IsLocked=0, resets FailedAttempts to 0
sp_Users_GetAll	SELECT	Returns all users with filter by IsActive, IsLocked
sp_Users_GetByID	SELECT	Returns single user record by UserID
sp_Users_GetByUsername	SELECT	Returns user for login — used by auth service
sp_UserRoles_Assign	INSERT	Assigns a role to a user — ignores if already assigned
sp_UserRoles_Revoke	DELETE	Removes a role assignment from a user
sp_UserRoles_GetByUser	SELECT	Returns all roles assigned to a given UserID
sp_Users_GetEffectivePerms	SELECT	Returns merged permission set for a user across all roles

14.4 OTP

Procedure Name	Action	Description
sp_Users_OTP_Generate	INSERT	Generates and stores OTP + expiry; called when IsOTPEnabled=1
sp_Users_OTP_Validate	SELECT	Validates submitted OTP against stored value and expiry

14.5 Organisation and School

Procedure Name	Action	Description
sp_Orgs_Create	INSERT	Creates a group or school record in tbl_mst_organisations
sp_Orgs_Update	UPDATE	Updates organisation record — writes to history table
sp_Orgs_GetAll	SELECT	Returns all organisations, filterable by type (Group/School)
sp_Orgs_GetByID	SELECT	Returns a single organisation by OrgID
sp_UserSchools_Assign	INSERT	Assigns a school/branch to a user
sp_UserSchools_Revoke	DELETE	Removes a school/branch assignment from a user
sp_UserSchools_GetByUser	SELECT	Returns all schools assigned to a given UserID

14.6 Department and Menu

Procedure Name	Action	Description
sp_Dept_Create	INSERT	Creates a department record
sp_Dept_Update	UPDATE	Updates department — writes history
sp_Dept_GetAll	SELECT	Returns all departments for the active organisation
sp_UserDept_Assign	INSERT	Assigns a department to a user
sp_MenuItems_GetByRole	SELECT	Returns dynamic menu items for a given RoleID

14.7 Audit Log

Procedure Name	Action	Description
sp_AuditLog_Insert	INSERT	Writes a system-level audit log entry (login/logout/profile switch)
sp_AuditLog_GetByUser	SELECT	Returns audit entries filtered by UserID and date range
sp_AuditLog_GetByModule	SELECT	Returns audit entries filtered by ModuleName and date range

19 Authentication, Authorization and Security

Control Area	Implementation
Authentication	JWT Bearer token authentication. On login, sp_Users_GetByUsername verifies credentials and returns signed JWT with UserID and RoleIDs. If OTP is enabled (IsOTPEnabled=1), OTP verification is required as second factor before JWT is issued.
OTP Flow	OTP is configurable at user creation (IsOTPEnabled flag). When enabled: (1) User submits ID + Password; (2) System validates credentials; (3) OTP sent via configured channel; (4) User submits OTP; (5) JWT issued on success. ID + Password is always required — OTP is additional.
Authorization	Claims-based RBAC on every request. Middleware reads JWT, loads vw_UserEffectivePerms, checks ModuleCode+Action pair.
Password Security	AES-256 hashing with per-user salt. PasswordHash stored in tbl_mst_users. Plain-text password never stored or logged.
Account Lockout	5 failed attempts trigger IsLocked=1. Unlock requires Super Admin action via sp_Users_Unlock.
Input Security	Model validation attributes, anti-forgery tokens (CSRF), server-side validation, parameterized queries in all stored procedures.
Transport Security	HTTPS enforced. Secure and HttpOnly cookie flags. HSTS enabled in production.
Audit & Traceability	All create/update/delete/assign/revoke events captured in per-table history tables (UpdateDeleteFlag, DoneBy, DoneOn). System-level events (login, logout, profile switch) go to tbl_aud_log.

18 Business Rules and Validations

Rule	Detail
Unique Role Name	RoleName must be unique across all active roles. Validated server-side via sp_Roles_Create.
Unique Username & Email	Username and Email must each be unique. Enforced via UQ indexes + service-layer validation.
No Direct Permissions	Permissions cannot be assigned directly to users. All access must flow through a Role.
Role Deactivation Check	Roles with active user assignments cannot be deactivated. sp_Roles_Deactivate checks first.
Soft Delete Only	No hard deletes permitted. IsActive=0 used for all deactivation. Triggers guard against direct SQL deletes.
Multi-Role Union	A user with multiple roles gets the UNION of all role permissions. No conflict resolution needed.
Password Policy	Minimum 8 chars, upper + lower + digit + special character. AES-256 hashed with per-user salt.
Failed Login Lock	After 5 consecutive failed attempts IsLocked=1. Only Super Admin can unlock via sp_Users_Unlock.
Mandatory Audit Logging	Every CREATE / UPDATE / DELETE / ASSIGN / REVOKE must be captured in per-table history tables (UpdateDeleteFlag, DoneBy, DoneOn). LOGIN / LOGOUT / PROFILE_SWITCH go to tbl_aud_log.
Mandatory Field Validation	All NOT NULL fields validated client-side (JS) and server-side (model validation + stored procedure checks).
No Ad-hoc SQL	Even minor data fetch must go through a stored procedure. ADO.NET only. No ORM. No inline SQL in application code.
Back-date Control	AllowBackdate flag on tbl_mst_users controls whether the user can enter data for past dates. Enforced at service layer.
School-scoped Data	All queries include MainAccountID + active OrgID. Users cannot see data outside their assigned schools.
SRS Approval Gate	Development proceeds only after client approval of the SRS. Any future changes are documented separately.

27 Notes on Remaining Sections

Sections 05 (System Architecture), 07 (Functional Flow), 08 (Role & Permission Hierarchy), 09 (Module-Wise Technical Design), 10 (Project Structure), 12 (Entity Relationship Diagram), 13 (Index Details), 15 (Views), 16 (Triggers), 17 (RBAC Authorization Flow), 20 (API Reference), 21 (Configuration Management), 22 (Logging, Error Handling and Monitoring), 23 (Performance and Scalability), 24 (Deployment Architecture), 25 (Backup, Recovery, Audit Trail), 26 (Reports, Schedulers and Limitations) are carried forward from v1.2 without structural change. Updates to the ER diagram, index list, and trigger definitions to reflect new tables will be completed by the DBA team.